Suppression of the quadrupole-misalignment false EDM signal by orbit-based methods in a proton EDM storage ring

S. Hacıömeroğlu[1]

[1] *Affiliation*

(Dated: July 13, 2026)

In the storage ring proposed for measuring the proton electric dipole moment (EDM), quadrupole misalignments of a few microns generate, through a geometric (Berry) phase, a vertical spin-precession signal that mimics a genuine EDM and exceeds the target sensitivity of $10^{-29}\,e$cm by up to three orders of magnitude. Using symplectic spin tracking, we examine how far this false signal can be removed by measuring and correcting the beam orbit. We find that standard orbit correction alone does not guarantee suppression: the misalignment pattern separates into an orbit-visible anti-symmetric component and a symmetric component to which the orbit response is nearly blind (two orders of magnitude weaker in this lattice), and the false EDM is a bilinear functional of the two transverse misalignment patterns, so that it cannot be characterized by a single rms alignment tolerance. Null-finding beam-based alignment reaches the symmetric component that response-matrix inversion cannot; combined with orbit correction, it suppresses the false EDM from several hundred times the target to the target level in our simulations. We quantify the remaining requirements: the fidelity of the steering optics model, the re-alignment cadence against alignment drift, and the stability of the magnetic centers under gradient modulation, which is not yet modeled.

I. INTRODUCTION

A. The frozen-spin method and the false-EDM problem

A permanent electric dipole moment (EDM) of the proton would violate time-reversal symmetry, and its observation above the Standard-Model expectation ($\lesssim 10^{-31}\,e$cm) would constitute direct evidence of new sources of CP violation. The proposed storage-ring experiment [1] targets a sensitivity of $d = 10^{-29}\,e$cm using the *frozen-spin* method: protons are stored at the "magic" momentum ($p \approx 0.7$ GeV/c), at which the spin remains aligned with the momentum as the beam circulates. An EDM then reveals itself as a slow rotation of the spin out of the ring plane, driven by the radial electric bending field. The measured quantity is the out-of-plane precession rate,

$$f \equiv \frac{dS_y}{dt}, \qquad d = 10^{-29}\,e\cdot\text{cm} \;\leftrightarrow\; f \approx 1\ \text{nrad/s}. \tag{1}$$

Any competing effect that rotates the spin out of plane at this level must be eliminated or bounded; such effects are referred to as *false EDM* signals.

The subject of this paper is the false EDM generated by transverse misalignments of the focusing quadrupoles. A quadrupole has zero field on its magnetic axis, so a beam passing off-axis experiences a field proportional to the offset. A vertical offset dy exposes the beam to a radial magnetic field, which precesses the spin about the radial axis and can therefore tilt it out of the ring plane. A horizontal offset dx exposes the beam to a vertical field, which precesses the spin about the vertical axis; this precession is confined to the ring plane and, like the in-plane precession of the frozen-spin condition itself, does not accumulate in S_y. Either precession alone is nearly harmless: averaged around the ring, its out-of-plane effect largely cancels. When both are present, however, the two precessions do not commute — precessing first about the radial axis and then about the vertical axis is not the same operation as the reverse — and the residue of this non-commutativity accumulates turn after turn as a net rotation about the third axis, which is precisely the out-of-plane direction in which the EDM is sought. This is a geometric (Berry) phase, and its leading dependence is

$$f_{\text{false}} \propto dx \cdot dy \implies f_{\text{false}} \propto \sigma^2 \tag{2}$$

for misalignments of rms size σ. We verify the quadratic scaling directly in Sec. II B (measured exponent $p = 2.00 \pm 0.01$, in agreement with Fig. 9(a) of Ref. [1]). At a realistic alignment of $\sigma = 10\ \mu$m the false EDM is of order a thousand times the target of Eq. (1), and it appears in the same observable, dS_y/dt, as the genuine signal.

B. The ring

The lattice studied here is the symmetric-hybrid ring of Ref. [1]: electric bending with magnetic focusing. The ring ($R_0 = 95.49$ m) consists of 24 identical FODO cells. Each cell contains a horizontally focusing quadrupole (QF) and a horizontally defocusing quadrupole (QD) — both *magnetic*, with gradients of equal magnitude ($g \approx 0.2$ T/m) and opposite sign — separated by field-free drifts and by the *electric* bending deflectors:

$$\begin{gathered}\text{QF} \to \text{drift} \to \text{deflector} \to \text{drift} \to \text{QD}\\ \to \text{drift} \to \text{deflector} \to \text{drift} \to (\text{QF} \ldots)\end{gathered}$$

The deflectors are cylindrical electrostatic capacitors that bend the beam in the horizontal plane; at the magic momentum their radial field also freezes the in-plane spin precession. Two properties of this arrangement matter for what follows. First, between any two neighboring quadrupoles the beam traverses a deflector: the deflectors bend and weakly focus the beam *horizontally*, while in the *vertical* plane they act, to a very good approximation, as field-free drifts. Second, the alternating QF/QD gradients make the two transverse planes behave identically in the idealized lattice, with betatron tunes $Q_x \approx Q_y \approx 2.3$ (the tune is the number of transverse oscillations per revolution).

C. Orbit-based alignment as the natural remedy

Since the false EDM originates in the beam–quadrupole offsets, the natural strategy is to measure and correct them using the beam itself. This is standard accelerator practice. The closed orbit — the periodic trajectory about which the stored beam oscillates — is read continuously by beam position monitors (BPMs), located in our model at each of the 48 quadrupoles. Misaligned quadrupoles deflect the closed orbit, and the deflections are related to the offsets through the orbit response matrix; correcting the orbit with steering elements, guided by the singular-value decomposition of that matrix, has been routine since the work of Chung, Decker, and Evans [3]. Beyond orbit correction, the accelerator community locates individual quadrupole centers by *beam-based alignment* (BBA): the quadrupole gradient is varied and the beam position at which the orbit stops responding marks the magnetic center, a procedure that has been accelerated with ac excitation [5] and parallelized over many magnets [6]. Related tools include orbit-response modeling in the presence of degeneracy [4], correction schemes exploiting lattice symmetry [7], and on-line response-matrix refinement [8]. None of the measurement techniques examined in this paper is new in itself; the question we address is how they perform on the specific misalignment component that drives the false EDM — a question that, to our knowledge, has not been examined, and whose answer turns out to be structured.

D. Findings and outline

Three results organize the paper.

First, *standard orbit correction alone does not guarantee suppression of the false EDM* (Sec. III). The misalignment pattern separates into two components with sharply different orbit visibility: an *antisymmetric* component (neighboring QF and QD displaced oppositely), which produces a large orbit distortion and is removed by orbit correction, and a *symmetric* component (QF and QD displaced together), whose orbit signature is suppressed by roughly two orders of magnitude and which therefore survives correction. The false EDM is a bilinear functional of the horizontal and vertical misalignment patterns, and the surviving symmetric–symmetric term sets the post-correction floor. An immediate corollary is that the false EDM cannot be characterized by a single rms alignment tolerance: at fixed σ its value varies by more than an order of magnitude with the symmetric/antisymmetric content of the pattern.

Second, *the symmetric component cannot be recovered from orbit readings by response-matrix inversion or amplitude-readout techniques* (Sec. IV), for reasons that are coherent and therefore independent of BPM technology. This includes the counter-rotating beam separation proposed as a control observable in Ref. [1].

Third, *null-finding BBA, followed by a final orbit correction, does reach the symmetric component* (Sec. V). BBA locates each magnetic center by a zero-crossing rather than by inverting a matrix, and is therefore not subject to the visibility gap; the final orbit correction removes the antisymmetric residue that BBA leaves behind. In our tracking simulations the combination suppresses the false EDM from several hundred times the target to the target level, where each operation alone stalls an order of magnitude or more above it. Section VI examines the operational side — alignment drift, counter-rotating beams, and the time budget — and Sec. VII collects the conclusions. The simulation infrastructure and the spin-signal estimator, on which all quantitative statements rest, are described first in Sec. II.

II. SIMULATION METHOD

A. Tracking tools

All results are obtained with a particle-and-spin tracking code that integrates the equations of motion element by element through the fields of the lattice of Sec. I B, using a fourth-order Gauss–Legendre symplectic integrator; the spin is propagated in the same steps by the Thomas-BMT equation. Symplectic integration prevents the slow accumulation of energy and amplitude errors, which is a prerequisite for resolving precession rates at the nrad/s level over millions of turns. The genuine EDM enters the model through a switch; with the coupling set to $\eta = 1.88 \times 10^{-15}$ the code reproduces $dS_y/dt = 9.81 \times 10^{-10}$ rad/s, odd under beam reversal, in agreement with Eq. (C1) of Ref. [1]. All false-EDM results below are obtained with the EDM switch off, so that any measured out-of-plane precession is by construction a systematic effect. Quadrupole misalignments, gradient errors (β-beating), and roll angles enter the tracker directly as element parameters.

Wide parameter scans (noise Monte Carlo, β-beating sweeps) use a second, independent layer: the lattice functions, tunes, and the orbit response matrix,

$$R_{ij} = \frac{\sqrt{\beta_i \beta_j}}{2 \sin \pi Q} \, (KL)_j \cos\big(|\phi_i - \phi_j| - \pi Q\big), \qquad (3)$$

evaluated in closed form from the quadrupole and drift transfer matrices. This analytic layer runs in seconds, agrees with the tracker at the percent level in the vertical plane, and shares nothing with it except the configuration file, so that agreement between the two cannot arise from a common implementation error. Two of its limitations play a role later and are recorded here: it treats the electric deflectors as field-free drifts, which is accurate vertically but not horizontally (Sec. V); and an idealized response matrix can misrepresent the least-visible modes of the real beam dynamics, so response matrices used in inverse problems are always rebuilt from tracking.

B. Measuring the false EDM

The quantity of interest is the secular slope of $S_y(t)$. Extracting it reliably requires care, because the betatron oscillation of the tracked particle feeds oscillatory components into the spin motion that exceed the sought slope by orders of magnitude. Two measures are combined.

First, the tracked particle is placed on the closed orbit, where no betatron oscillation is present. The closed orbit is the fixed point of the one-turn map M in the transverse phase space (x, x', y, y'); it is found by Newton iteration, i.e. by solving $M(v) = v$ through repeated linearized steps $v \to v - (M' - I)^{-1}[M(v) - v]$, where M' is the Jacobian of the map obtained numerically from the tracker. A few iterations suffice.

Second, the spin history is fit with the model $S_y(t) = a + bt + \sum_k [c_k \cos\omega_k t + d_k \sin\omega_k t]$, and only the secular slope b is retained; residual oscillatory content is absorbed by the harmonic terms rather than biasing the slope, as it would in a plain polynomial fit.

The estimator was validated in three independent ways: (i) scanning $\sigma = 2.5$–10 μm yields $|f| \propto \sigma^p$ with $p = 2.00 \pm 0.01$, the pure quadratic scaling of a geometric phase with no linear leakage; (ii) under sign flips of the misalignment pattern the measured f obeys $f(+dx, -dy) = -f(+dx, +dy)$ and $f(-dx, -dy) = +f(+dx, +dy)$ at the 10^{-3} level, confirming the bilinear character of Eq. (2); (iii) the average of four particles launched symmetrically around the closed orbit reproduces the single on-orbit result within 1%.[1]

III. WHY ORBIT CORRECTION ALONE IS NOT SUFFICIENT

A. The residual after standard correction

For random misalignments of $\sigma = 10$ μm in both planes, the raw false EDM is of order 10^3 times the target [Eq. (1)]; a representative pattern used throughout this paper gives 356×. Standard orbit correction — flattening the closed orbit with steering correctors — reduces the ensemble median by a factor of about eight; combined with the counter-rotating–beam difference discussed in Sec. VI, the residual is about 62× the target ($\approx 6\times10^{-28}$ $e$cm), scaling as σ^2 (Fig. 1). Orbit correction is thus effective but not sufficient, and the remainder of this section identifies what the residual consists of.

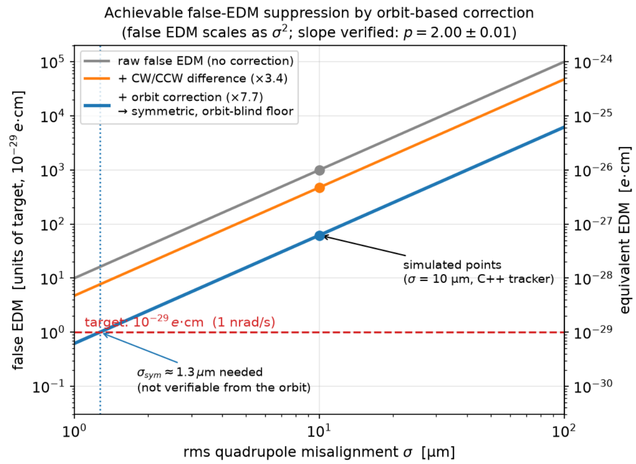

FIG. 1. False EDM before and after orbit-based correction versus the rms quadrupole misalignment. Each stage is anchored at the ensemble median of tracking results at $\sigma = 10$ μm (several random misalignment patterns per stage, not a single seed); all curves scale as σ^2 (verified exponent $p = 2.00$). Each curve should be read as the center of a band rather than a line: at fixed σ the false EDM varies by more than an order of magnitude with the symmetric/antisymmetric content of the pattern (Sec. III C), so the rms alone does not determine the signal.

B. Symmetric and antisymmetric misalignments

Let the 48 offsets of one plane form a vector v, with QF at index $2k$ and QD at index $2k{+}1$ in cell k. In each cell, the offsets of the two quadrupoles can be separated into the part in which they move *together* and the part in which they move *oppositely*,

$$
\begin{aligned}
(P_{\mathrm{sym}}\, v)[2k] &= (P_{\mathrm{sym}}\, v)[2k{+}1] = \tfrac{1}{2}\big(v[2k] + v[2k{+}1]\big),\\
(P_{\mathrm{anti}}\, v)[2k] &= -(P_{\mathrm{anti}}\, v)[2k{+}1] = \tfrac{1}{2}\big(v[2k] - v[2k{+}1]\big),
\end{aligned}
\tag{4}
$$

with $P_{\mathrm{sym}} + P_{\mathrm{anti}} = I$. The projectors act within one plane; they are applied independently to the horizontal and to the vertical offset vectors.

The physical distinction between the two components follows from the opposite-sign gradients of QF and QD. A misaligned quadrupole deflects the beam by an angle proportional to (gradient)×(offset). When two neighbors

[1] The symmetric four-particle average must be constructed around the closed orbit; constructed around the ideal axis it retains the common betatron component induced by the orbit offset itself and fails by orders of magnitude.

are displaced *oppositely* (antisymmetric component), the opposite gradients and opposite offsets combine to kicks of the *same* sign: the kicks add coherently around the ring into a slowly varying deflection pattern whose dominant azimuthal harmonics lie near the betatron tune, and the closed-orbit response is large. When two neighbors are displaced *together* (symmetric component), the kicks are opposite and cancel within each cell; what survives is a cell-to-cell alternation at azimuthal harmonic $k \approx 24$ (one sign change per cell, 24 cells). The closed-orbit response to a deflection pattern of harmonic k follows the gain law

$$G_k = \frac{C}{|Q^2 - k^2|}, \qquad C \approx 24.8,\ Q^2 \approx 5, \tag{5}$$

which is resonant for k near the tune and strongly suppressed far above it; with $Q \approx 2.3 \ll 24$ the symmetric component is nearly invisible [Fig. 2]. Quantitatively, a 10-μm-rms pattern leaves an orbit trace of ~ 115 μm if antisymmetric but only ~ 1.7 μm if symmetric.

The same conclusion can be read off the response matrix itself without reference to harmonics. Ranking the independent misalignment patterns by the strength of the orbit they produce (formally, by the singular values of R), the strongly-coupled patterns are found to be almost purely antisymmetric and the weakly-coupled ones almost purely symmetric; the overall visibility ratio between the strongest and weakest patterns is $\mathrm{cond}(R) = 193$ [Fig. 2, left]. Orbit correction, which by construction removes what the orbit sees, therefore removes the antisymmetric component and leaves the symmetric one.

C. The false EDM as a bilinear form: four channels

The connection between this decomposition and the false EDM can be stated in closed form. The closed orbit is linear in the offsets of each plane, $x_{\mathrm{CO}} = R_x\, v_x$ and $y_{\mathrm{CO}} = R_y\, v_y$. The geometric phase, at leading (second) order in the orbit, is the directed area swept by the spin, a quantity bilinear in the two transverse orbits:

$$f \propto \sum_j w_j \left[x_{\mathrm{CO}}\, y'_{\mathrm{CO}} - y_{\mathrm{CO}}\, x'_{\mathrm{CO}}\right]_{s_j} = x_{\mathrm{CO}}^{\top} K\, y_{\mathrm{CO}} \tag{6}$$

for a lattice-dependent kernel K. Substituting the orbit response,

$$f = v_x^{\top} \left(R_x^{\top} K R_y\right) v_y \;\equiv\; B(v_x, v_y). \tag{7}$$

Splitting each plane independently into its symmetric and antisymmetric parts, $v_x = v_x^{\mathrm{s}} + v_x^{\mathrm{a}}$ and $v_y = v_y^{\mathrm{s}} + v_y^{\mathrm{a}}$ [Eq. (4)], bilinearity separates f into four channels,

$$f = \underbrace{B(v_x^{\mathrm{s}}, v_y^{\mathrm{s}})}_{f_{ss}} + \underbrace{B(v_x^{\mathrm{s}}, v_y^{\mathrm{a}})}_{f_{sa}} + \underbrace{B(v_x^{\mathrm{a}}, v_y^{\mathrm{s}})}_{f_{as}} + \underbrace{B(v_x^{\mathrm{a}}, v_y^{\mathrm{a}})}_{f_{aa}}. \tag{8}$$

The two planes are coupled by the kernel B, not by the projectors. Since each factor $R\,v$ is large for an antisymmetric pattern and G_k-suppressed for a symmetric one, the pure-symmetric channel f_{ss} is by far the smallest, and every channel containing an antisymmetric factor is larger.

The channels are measured directly, and this measurement method underlies all symmetric/antisymmetric attributions in this paper: the misalignment vectors are projected with Eq. (4), each projected pair (v_x^a, v_y^b) is loaded separately into the tracker, and the resulting f is extracted with the estimator of Sec. II B. Figure 3 shows the result at $\sigma = 10$ μm: the four signed channels sum to the full false EDM within 0.1% ($f/\sum = 0.999$–1.001 over three seeds), confirming Eq. (8); the pure-symmetric channel is the smallest in every seed; and each channel individually scales as σ^2. The gain law accounts for the ordering but not fully for its magnitude: the orbit-trace ratio between antisymmetric and symmetric patterns of equal rms is ~ 70, while the measured channel ratios are one to two orders of magnitude, the difference arising because the kernel of Eq. (6) weights orbit *derivatives*, which partially compensates the suppression of the high-harmonic symmetric orbit. The measured channels, rather than the harmonic estimate, are therefore used as the quantitative reference throughout.

Two consequences frame the rest of the paper. First, the false EDM is not a function of the rms misalignment: it is σ^2-homogeneous, but at fixed σ its value depends on the pattern — in Fig. 3 the channels span $3.5\times$ to $900\times$ the target at the same 10 μm rms. A single alignment tolerance therefore cannot summarize the systematic, and a correction procedure changes not only the size but the *statistics* of the misalignment: it converts a random pattern into a structured one, concentrated in whichever component the procedure cannot see. Second, the division of labor for suppression is fixed by Eq. (8): orbit correction removes the antisymmetric factors and with them the three large channels; the remaining floor is f_{ss}, and any further progress requires a measurement that reaches the symmetric component itself.

IV. THE SYMMETRIC COMPONENT CANNOT BE RECOVERED BY INVERSION OR AMPLITUDE READOUT

Before presenting the measurement that succeeds, we summarize the techniques that do not, and why. All of them attempt to reconstruct the misalignments (or their symmetric part) from orbit readings. The realistic error model includes static BPM offsets (~ 100 μm each, the electronic zero of a monitor relative to the true axis), white BPM noise (~ 1 μm per reading, reducible by averaging), and a coherent optics-model error (β-beating, implemented as per-quadrupole gradient errors of 0.5–5%; a machine calibrated with standard LOCO techniques typically retains ~ 1%).

Difference-orbit inversion. Modulating all quadrupole gradients between two settings and subtracting the orbits cancels the static BPM offsets exactly and leaves a linear

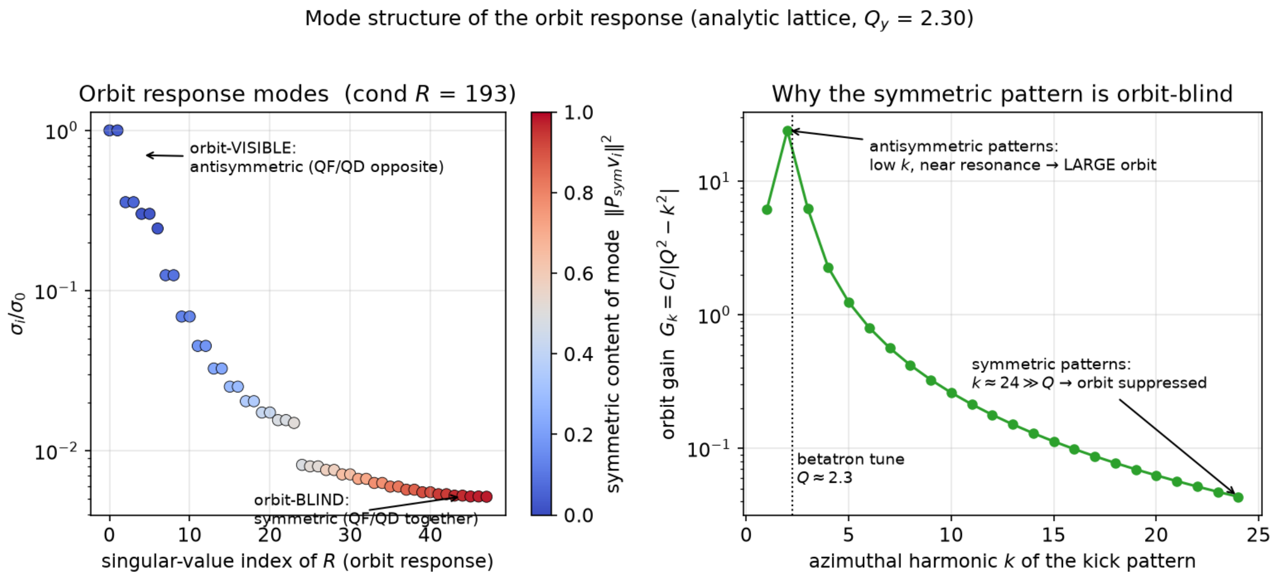

FIG. 2. Orbit visibility of misalignment patterns (analytic lattice, $Q_y = 2.30$). Left: independent misalignment patterns ranked by the strength of their orbit response; patterns that couple strongly to the orbit are antisymmetric (QF, QD opposite), patterns that couple weakly are symmetric (QF, QD together); the visibility ratio is cond$(R) = 193$. Right: the orbit gain law of Eq. (5); antisymmetric patterns drive low harmonics near the betatron resonance, symmetric patterns drive $k \approx 24$, far above it.

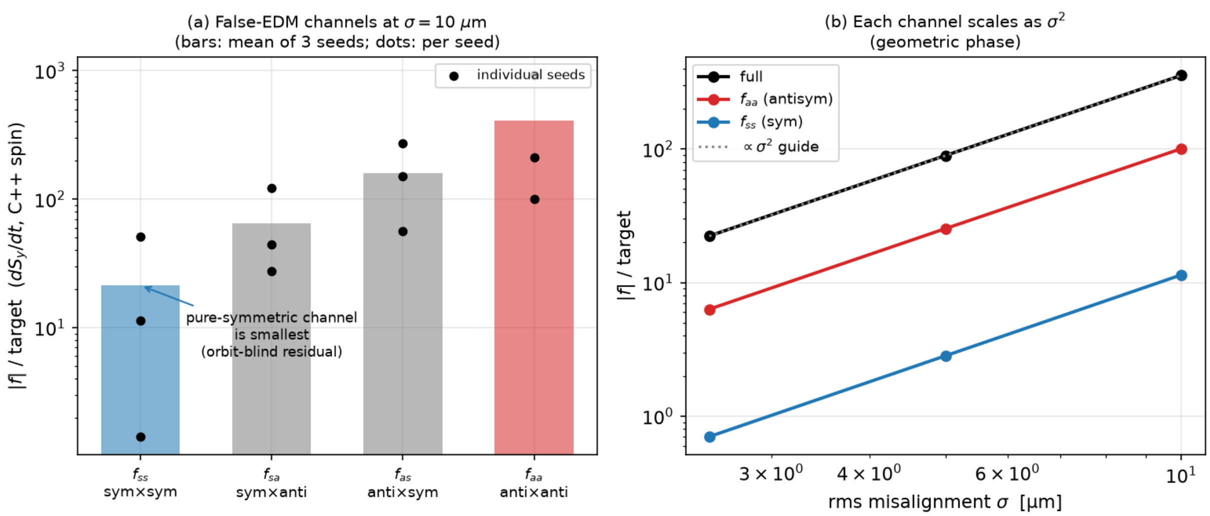

FIG. 3. The four false-EDM channels of Eq. (8), measured by projecting the misalignments and tracking each part separately. (a) At $\sigma = 10$ μm the pure-symmetric channel f_{ss} is the smallest, and every channel containing an antisymmetric factor is larger (bars: mean of three seeds; dots: individual seeds; the spread reflects the signed, bilinear character of f). (b) Each channel scales as σ^2. The four signed channels sum to the full signal within 0.1%.

system $\Delta y = \Delta R\,\Delta q$. With no errors the inversion is exact. Its weakness is conditioning: cond$(\Delta R) = 3.7 \times 10^4$, with the weak directions being precisely the symmetric ones, so reconstruction errors are amplified along the component of interest. White noise can be beaten by synchronous detection — modulating at ~ 1 kHz and averaging reaches an effective noise of ~ 14 nm within minutes — but the optics-model error cannot: it is coherent, identical in every average, and at β-beating of only 0.5% the reconstructed symmetric component is several times larger than the true one (283 μm against a 41-μm signal in our standard test) while the antisymmetric component remains accurate [Fig. 4]. A caution for the evaluation of such reconstructions: the Pearson correlation between reconstruction and truth, dominated by the well-reconstructed antisymmetric components, re-

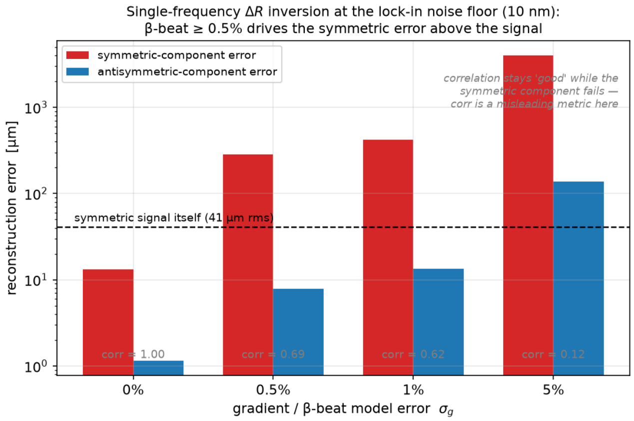

FIG. 4. Difference-orbit inversion at the synchronous-detection noise floor (10 nm, 40 seeds): reconstruction error of the symmetric and antisymmetric components versus the optics-model error. The dashed line marks the size of the symmetric signal itself. A model error at the realistic $\geq 0.5\%$ level drives the symmetric error far above the signal while the Pearson correlation (gray labels) remains high.

mains as high as 0.7–0.99 in these failures; the meaningful metric is the rms error of each component separately, compared with the size of that component in the signal. Because the limit is coherent, it is independent of BPM technology; lower-noise pickups, including SQUID-based ones, shorten the integration time but do not move the floor. The same structural point can be stated as a bound: any estimator that cancels static offsets by a two-setting difference pays for the cancellation with a $\|\Delta R^{-1}\| \sim \|R^{-1}\|/\varepsilon_g$ amplification, where ε_g is the relative setting difference.

Per-quadrupole modulation read as an amplitude. Assigning each quadrupole its own modulation frequency and reading the per-frequency amplitude at the BPMs — the ac-BBA concept of Ref. [5] — promises a direct, inversion-free readout: a quadrupole whose center is offset from the beam by o_i imprints, when its gradient is modulated, a dipole-like deflection proportional to o_i (the *feed-down* kick, the dipole field component seen by a beam passing off-center). The method fails in this ring for an instructive reason [Fig. 5]. Modulating the gradient of one quadrupole modulates two things at once: the feed-down kick of that quadrupole (the sought signal, ~ 0.9 μm at a BPM), and the focusing of the ring as a whole — the lattice functions and the tune change slightly, so the *entire pre-existing closed orbit*, built by all 48 offsets together and much larger ($\sim$0.37 mm rms) than any single feed-down kick, is re-transported through slightly different optics. This global response, which we refer to as optics *breathing*, appears at the same modulation frequency as the signal and exceeds it about sevenfold; a decomposition test shows the demodulated amplitude to be dominated by the neighbors' orbit rather than the quadrupole's own offset by a factor of 266. Because breathing is coherent with the modulation, it does not average away with more BPMs or lower noise; removing it in an idealized model restores perfect reconstruction, which pins the failure entirely on this mechanism.

Learned and structured estimators. Because the misalignment-to-orbit map is linear, a neural network trained on simulated orbit data converges to the regularized pseudo-inverse of R and inherits its conditioning; in our tests a multilayer perceptron matched truncated-SVD inversion on both components (symmetric error 5.6 vs. 6.3 μm against a 9.9-μm signal) and improved on it nowhere. In this linear, noise-limited setting the limitation is the information content of the data, not the expressiveness of the estimator, though we note that this argument does not extend to settings with useful nonlinearity or strong priors. Sparsity-based reconstructions (LASSO, CLEAN, greedy harmonic selection) reach the same floor. Restricting the reconstruction basis to *a priori known* harmonics improves the conditioning by two orders of magnitude and works — but which harmonics are present is precisely what is unknown; and fitting a known signature (for example a k=2 stray-field imprint) is well-conditioned but biased, since one harmonic coefficient lumps together the field, the misalignment content, and the BPM-offset content at that harmonic. A calibrate-then-monitor strategy (LOCO followed by drift monitoring of the difference orbit) tracks *changes* of the antisymmetric component well — 6.5 μm rms under 50 μm BPM offsets — but the difference orbit passes through the same response matrix and is equally blind to the symmetric component.

Counter-rotating beams, polarity switching, and the tune. The control chain of Ref. [1] monitors the *separation* of the two counter-rotating (CR) closed orbits and reduces it with correctors. The CR separation, however, is itself an orbit observable and inherits the orbit's visibility structure: in our tracking the antisymmetric-to-symmetric signature ratio is 5.7× in a single-beam orbit and 8.0× in the CR separation (an independent pattern set gives 3.8× and 4.5×) [Fig. 6] — the CR separation opens no additional window on the symmetric component. A related question is why the beam-reversal comparison does not cancel the geometric phase outright. In the idealized periodic lattice we measure the reversed beam to be exactly equivalent to the forward beam with all quadrupole polarities flipped (ratio 1.00); and the polarity flip is *not* a symmetry of the geometric phase, because the phase is an ordered accumulation of non-commuting precessions — reversing the traversal order is not equivalent to reversing the sign. Measured flip ratios $f(-g)/f(+g)$ of +2.8, +0.85, and +1.17 for symmetric, antisymmetric, and generic patterns confirm that the phase is approximately even in the gradient but with pattern-dependent deviations; consequently the CW/CCW difference cancels a pattern-dependent fraction of the false EDM — a factor ~ 3 in practice (Sec. VI) — rather than all of it, and polarity switching provides no independent cancellation knob for the

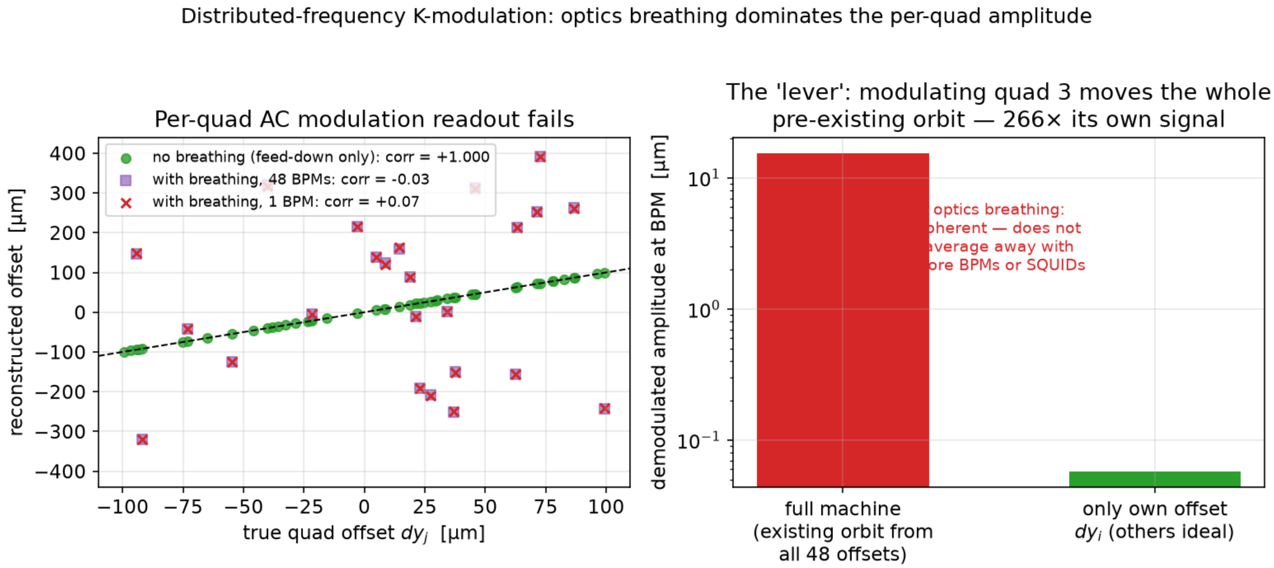

FIG. 5. Per-quadrupole gradient modulation read as an amplitude. Left: reconstructed versus true offsets — exact when the optics breathing is artificially removed (green); uncorrelated with it present, whether one BPM (red) or all 48 (purple) are used. Right: the demodulated amplitude responds to the re-transported pre-existing orbit at 266× the level of the quadrupole's own feed-down signal; the contamination is coherent and does not average down.

quadrupole-alignment geometric phase (it remains effective against systematics that are odd in the gradient). Finally, raising the betatron tune toward the symmetric kick harmonic would make the symmetric component visible in principle; in practice the stopband limit (180° phase advance per cell) caps the tune at $Q \leq 12$, recovering only the fast-varying part of the symmetric family (its high azimuthal harmonics m, where m indexes the harmonics *within* the 24-dimensional symmetric subspace, the slowly-varying members requiring $Q \geq 16$), and the stronger gradients required ($0.21 \to 0.69$ T/m for $Q_y : 2.3 \to 10.3$) raise the false EDM itself steeply ($\sim g^3$, a measured factor of 32) while the genuine EDM signal, set by the electric field, is unchanged. A further alternative outside the orbit toolbox — operating slightly off the magic momentum to sense the geometric phase resonantly — fails for reasons detailed in Appendix A.

The common conclusion of this section is structural. For every technique in the inversion and amplitude-readout families, the limiting effect is coherent — conditioning amplified by β-beating, or optics breathing — and coherent effects do not average down. *Within the simulated model, no BPM technology — including SQUID-based BPMs — changes these floors.* We emphasize the scope of this statement. It is a structural property of the estimator classes considered, established by class-level arguments (the conditioning bound and the coherence of the breathing term) together with systematic simulation; it is not an information-theoretic impossibility claim for all conceivable orbit-based measurements. Indeed, the next section exhibits an orbit-based measurement outside these classes that does reach the symmetric component.

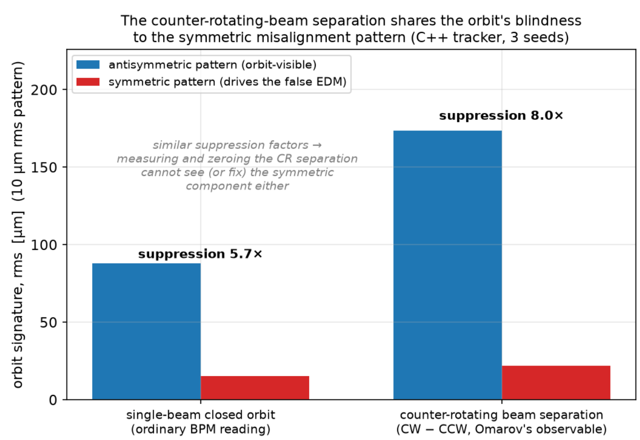

FIG. 6. The counter-rotating beam separation (right group) suppresses the symmetric pattern by a factor similar to the ordinary single-beam orbit (left group): as an orbit observable it shares the orbit's blindness to the component that drives the residual false EDM.

V. REACHING THE SYMMETRIC COMPONENT: NULL-FINDING BBA

A. Procedure

Standard quadrupole beam-based alignment does not invert a matrix and does not read an amplitude against a calibration; it locates each magnetic center as the beam position at which the response to a gradient change *vanishes*. For each of the 47 alignable quadrupoles in turn:[2] (i) using the nearest steering corrector, scaled by the single response-matrix element R_{ij}, the beam is placed at two positions bracketing the expected center of quadrupole i; (ii) at each position the gradient of quadrupole i is changed by a small amount and the difference of the two closed orbits is recorded at all 48 BPMs — this difference is the feed-down imprint, proportional to the beam–center offset; (iii) since the imprint changes sign as the beam crosses the center, the center is obtained as the weighted least-squares zero over the BPMs, with the spread of the per-BPM zero crossings retained as a per-magnet quality measure. All orbits are read with the beam placed on the 4D closed orbit (Sec. II B).

The role of the response matrix in this procedure is confined to the *forward* direction: a single, well-conditioned element sizes the steering bump. No global inverse appears anywhere, so the conditioning wall of Sec. IV does not apply; the 47 centers are measured locally and independently, and the symmetric combination of the answers is determined as well as any other combination. An error in the forward element mis-sizes a bump and biases one local measurement — an effect that will be visible below — but it does not re-introduce the collective visibility gap.

B. Idealized and realistic optics

In idealized optics the procedure works outright: the symmetric centers are recovered, and the false EDM of the representative pattern drops from 356× to 1.6× the target.

Under a realistic 1% β-beating a single pass fails, through the same breathing mechanism as in Sec. IV: modulating one quadrupole re-transports the large uncorrected orbit through the beating optics and biases the zero crossing. The remedy is iteration — correct the orbit with the current estimates, then re-measure — since a smaller orbit breathes less. Iterating with under-relaxation and the per-magnet quality cut, the vertical plane converges cleanly (residual halving each pass, reaching 0.9 μm). The horizontal plane initially stalled, and the diagnosis is instructive. The steering bumps were sized with the analytic optics model, which treats the deflectors as drifts (Sec. II A). Vertically this is accurate, and indeed the model reproduces the tracker's vertical tune to four decimal places (fractional tune 0.3032 in both). Horizontally the deflectors focus and bend, and the model does not know it: it predicts equal tunes in the two planes, while the tracker's horizontal fractional tune is higher by 0.052; element by element, the model's horizontal response matrix is essentially uncorrelated with the measured one (median ratio −0.05), and it selects the wrong steering corrector for 36 of the 47 magnets, against a 5%-accurate response and 12 wrong selections vertically. Re-building the steering from the *measured* response matrix — ordinary practice on a real machine — repairs the horizontal plane: the per-magnet quality ceases to throttle the horizontal corrections, and the false EDM after three passes improves ninefold over the analytic-model steering (Table I).

TABLE I. Iterated null-finding BBA under 1% β-beating (tracking, three passes), steering with the idealized analytic model versus the measured response matrix. "sym dx/dy" are the symmetric residuals of the measured-matrix run. Both runs solve the vertical plane; only the measured matrix also corrects the horizontal one.

pass	sym dy [μm]	sym dx [μm]	$\lvert f\rvert$ / target analytic	measured
raw	—	—	356×	356×
1	3.27	4.27	368×	73×
2	1.70	2.90	238×	**17×**
3	0.91	2.29	145×	28×

C. Closing the chain: a final orbit correction

The 17–28× residual of the measured-matrix run is not the symmetric floor. Decomposing the third-pass residual with the channel method of Sec. III C: the symmetric part contributes 3.5× the target, the antisymmetric part — a ∼2 μm leftover of the still-imperfect horizontal centering — contributes 16×, and cross terms account for the remainder, in the ordering expected from Eq. (8). The antisymmetric leftover is orbit-visible, and a single standard orbit correction removes it: applied to the BBA residual, it brings the single-beam false EDM from 28× to 0.03–0.8× the target, depending on the correction depth. Scaling the corrected residual down confirms a clean σ^2 dependence with no additive floor over the tested range.

The resulting picture is consistent and complementary: *orbit correction removes the antisymmetric channels of Eq. (8); null-finding BBA reduces the symmetric channel that orbit correction cannot see; in combination they reach the target, while each alone stalls* (62× for orbit correction, ∼20× for BBA without the final correction). Table II summarizes the chain. Two caveats bound the claim. The sub-target endpoint is a single misalignment seed of a signed, bilinear quantity whose value scatters from seed to seed (Fig. 3); the robust statement is suppression to the symmetric floor of a few times the target, and a multi-seed campaign to fix the distribution is in progress. And the entire analysis assumes that gradient modulation does not itself move the magnetic centers;

[2] The gradient-modulated quadrupole of cell 0 is a distinct element type in our model and is excluded; its offset is set to zero and reported as such.

TABLE II. The suppression chain at $\sigma = 10\ \mu$m (single beam, tracking). The second row is an ensemble median including the counter-rotating–beam difference; the remaining rows follow the representative pattern (raw 356×). The equivalent EDM uses 1 nrad/s $\leftrightarrow 10^{-29}$ $e$cm. The final entry is a single-seed result; the multi-seed distribution is under study.

Stage	$\lvert f\rvert$ / target	EDM [$e$cm]
raw	356×	$\sim 4 \times 10^{-27}$
orbit corr. + CW/CCW	$\sim 62\times$	$\sim 6 \times 10^{-28}$
BBA (measured steering)	17–28×	$\sim 2 \times 10^{-28}$
BBA + final orbit corr.	0.03–0.8×	$\lesssim 10^{-29}$

hysteresis or thermal effects at the micron level would enter the measurement directly, and this hardware floor has not been modeled.

VI. OPERATING THE SCHEME: DRIFT, COUNTER-ROTATING BEAMS, AND THE TIME BUDGET

An alignment campaign is useful only if its result survives on the machine. Quadrupole positions drift — thermally, with ground motion, and with magnetic history — so the corrected state of Sec. V is a snapshot, onto which the machine continuously re-loads random misalignment. We simulated this directly by adding fresh random offsets of rms $\sigma_{\rm d}$ to the corrected state and re-measuring the false EDM, with and without orbit feedback.

Without any re-correction the false EDM returns quickly: $\sigma_{\rm d} = 10\ \mu$m restores hundreds of times the target. With *continuous orbit correction* — routine, fast, and non-invasive on any storage ring — the picture changes qualitatively, because random drift is shared between the two components and the orbit feedback continuously removes its antisymmetric part: the false EDM stays below the target for $\sigma_{\rm d}$ up to about 5 μm (0.05× at 2 μm, 0.3–0.6× at 5 μm) and reaches only a few times the target at 10 μm. Only the slowly accumulating *symmetric* part of the drift requires a repetition of the BBA campaign. The alignment therefore does not need to be *held* at the micron level mechanically; it needs to be re-established at a cadence set by the symmetric drift rate.

The counter-rotating beam difference adds a further, independent factor. Applied to the corrected state, the CW/CCW combination suppresses the residual false EDM by an additional factor of 2–3 in our single-seed tests, consistent with the pattern-dependent parity discussed in Sec. IV; an ensemble determination of this factor, using the full sign bookkeeping of Ref. [1], is left for the multi-seed campaign.

These sensitivities translate into a time budget whose absolute numbers require one machine-specific input that simulation cannot supply: the drift rate. The initial BBA campaign (47 magnets × 2 planes, iterated) costs hours to a day of dedicated beam time. Continuous orbit correction runs during data taking. If the total random drift is of order 1 μm/day — a representative figure for a thermally stabilized installation, to be confirmed for the actual site — its symmetric part reaches the 2 μm scale in roughly three days, setting the BBA repetition cadence; at hours per campaign this is a duty-cycle loss at the percent level. The dominant time scale of the experiment remains the statistical one: polarimetry reaches the nrad/s level only over months to a year of running [10], and the alignment maintenance described here fits inside that schedule provided the drift rate is not far above the representative figure. The unmodeled hardware floor — magnetic-center motion under gradient modulation — enters this budget as a repeatability limit of each BBA campaign and is the main open item on the operational side.

VII. CONCLUSIONS

The quadrupole-misalignment false EDM of the symmetric-hybrid proton EDM ring is controlled by the interplay of two misalignment components with sharply different orbit visibility. The central conceptual result of this work is the bilinear four-channel structure of Eq. (8), which explains both why standard orbit correction stalls and which measurement completes it; the quantitative picture is as follows.

(1) Standard orbit-based control — orbit flattening together with the counter-rotating–beam difference — suppresses the false EDM by roughly an order of magnitude and no further: it removes the orbit-visible, antisymmetric component of the misalignment, and the weakly observable symmetric component sets a floor of $\sim 6 \times 10^{-28}$ $e$cm at $\sigma = 10\ \mu$m, scaling as σ^2.

(2) The false EDM is a bilinear functional of the two transverse misalignment patterns [Eq. (8)], verified in tracking at the 0.1% level. It is therefore not characterized by an rms tolerance: at fixed σ it varies by more than an order of magnitude with the pattern. More generally, a correction procedure changes the statistics of the misalignment — it converts random errors into a structured residual concentrated in the component the procedure cannot see — so the driving question is not how small the residual is, but in which component it lies.

(3) In our model, the symmetric component is unreachable from orbit readings by the inversion and amplitude-readout families of techniques, for coherent reasons (conditioning amplified by optics-model error; optics breathing) that improved BPM technology does not mitigate. The counter-rotating beam separation, being an orbit observable, shares this blindness; polarity switching provides no independent cancellation for the quadrupole-alignment geometric phase; and tune elevation is capped by the stopband well below the symmetric kick harmonic.

(4) Null-finding BBA is not subject to these limits, because it locates each magnetic center by a local zero crossing rather than through a global inverse. It requires

a faithful *horizontal* steering model — in this lattice the electric deflectors focus horizontally, and an idealized model that neglects this selects wrong correctors and stalls the horizontal plane — and it leaves an antisymmetric residue that a final orbit correction removes. The full chain, BBA with measured-matrix steering plus orbit correction, suppresses the single-beam false EDM from 356× to the target level in our simulations (single seed; multi-seed campaign in progress), with the CW/CCW difference contributing a further factor of 2–3. Two qualifications belong to this statement. The ninefold gain of the measured over the idealized steering matrix is a property of the fidelity of the optics model supplied to the procedure, not of the algorithm, which is standard; and the chain assumes throughout that the magnetic centers remain stationary under the gradient modulation itself.

(5) Operationally, the corrected state need not be held statically: continuous orbit correction absorbs random drift up to ~ 5 μm rms, and only the symmetric drift component requires periodic re-alignment, at a cadence compatible with the statistical time scale of the experiment for drift rates of order 1 μm/day.

The paper thus both diagnoses the limitation of orbit-based control and demonstrates its resolution at the proof-of-principle level; it is not yet a tolerance document. The principal open items are the magnetic-center stability under gradient modulation (the hardware floor of the BBA measurement, not yet modeled), the multi-seed distribution of the final residual, the ensemble determination of the CW/CCW factor, and the portability of the quantitative results to other lattices, which we have argued structurally but verified on one lattice only. All numbers refer to one lattice in a linear model without sextupoles; the structural conclusions — the visibility split, the bilinear channel decomposition, and the complementarity of orbit correction and BBA — follow from the alternating-gradient principle and are expected to carry over to other strong-focusing implementations.

ACKNOWLEDGMENTS

Appendix A: Off-momentum spin sensing

For completeness we document an attempted escape outside the orbit toolbox: sensing the geometric phase with the spin itself by storing the beam slightly off the magic momentum. Away from the magic momentum the in-plane spin precession unfreezes and the spin precesses at a tune $\nu_s \neq 0$; the geometric-phase contribution to the vertical spin component then appears as an oscillation whose amplitude is *enhanced* by $1/\nu_s$, suggesting a fast, resonant readout of the misalignment channel that could bypass the orbit entirely.

The idea fails against a first-order background. Misalignments not only generate the second-order geometric phase; they also tilt the stable spin direction (the invariant spin axis) at *first* order in σ. Off-magic, this tilt produces a vertical spin oscillation at the same frequency ν_s as the sought signal and exceeds it, at $\sigma = 10$ μm, by a factor of order 3600. We tested the available separation channels and found all closed: the two terms scale indistinguishably with the momentum offset; their relative phase is pattern-dependent (spread of 91° over seeds), so no fixed phase gate isolates the signal; symmetric four-particle sampling does not cancel the first-order term, which is common to the four particles; the amplitude crossing between the two terms lies at an unphysical $\sigma \approx 62$ mm; a full three-dimensional fit of the invariant axis absorbs the oscillation entirely, leaving no residual to read; and the longitudinal spin component is insensitive to the relevant knobs off-magic. The magic momentum is the unique operating point at which the dominant first-order effect is converted into a harmless constant offset — which is also why the estimator of Sec. II B functions there.

[1] Z. Omarov, H. Davoudiasl, S. Hacıömeroğlu, V. Lebedev, W. M. Morse, Y. K. Semertzidis, A. J. Silenko, E. J. Stephenson, and R. Suleiman, "Comprehensive symmetric-hybrid ring design for a proton EDM experiment at below 10^{-29} $e$cm," Phys. Rev. D **105**, 032001 (2022).

[2] S. Hacıömeroğlu and Y. K. Semertzidis, "Systematic errors related to quadrupole misplacement in an all-electric storage ring for proton EDM experiment," arXiv:1709.01208.

[3] Y. Chung, G. Decker, and K. Evans, "Closed orbit correction using singular value decomposition of the response matrix," in *Proceedings of PAC 1993*, p. 2263.

[4] S. Wegscheider and J. Vilsmeier *et al.*, "Inverse modeling of circular lattices via orbit response in the presence of degeneracy," Phys. Rev. Accel. Beams **26**, 032803 (2023).

[5] Z. Martí, G. Benedetti, U. Iriso, and A. Franchi, "Fast beam-based alignment using ac excitations," Phys. Rev. Accel. Beams **23**, 012802 (2020).

[6] X. Huang, "Simultaneous beam based alignment measurement for multiple magnets," arXiv:2203.14869.

[7] S. H. Mirza, R. Singh, P. Forck, and H. Klingbeil, "Closed orbit correction at synchrotrons for symmetric and near-symmetric lattices," Phys. Rev. Accel. Beams **22**, 072804 (2019).

[8] V. Ziemann and M. Ziemann, "Noninvasively improving the orbit response matrix while continuously correcting the orbit," arXiv:2104.05300.

[9] J. Rossbach, "Closed-orbit distortions of periodic FODO lattices due to plane ground waves," Part. Accel. **23**, 121 (1989).

[10] F. Müller *et al.* (JEDI Collaboration), "A new beam po-

larimeter at COSY to search for electric dipole moments of charged particles," arXiv:2010.13536.